CMPG 325 – Computer Networks
Milestone 3: Technical Report

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Client ID	CLI-084
Assigned Organisation	Ramothibe Labour Law Consultants (Kimberley)
Industry	Legal Services

1. Executive Summary

I designed and built a small multi-VLAN network for Ramothibe Labour Law Consultants, a legal services firm in Kimberley, using Cisco Packet Tracer and the assigned address block 172.30.54.0/23. The network has five VLANs, three switches joined by 802.1Q trunks, and a Cisco 1941 router that routes between the VLANs using router-on-a-stick. Voice traffic has its own VLAN to cover change request CR11 (VoIP handsets). My assigned feature, the static address 172.30.54.130 inside that Voice VLAN, works and was tested by pinging it from a host in the staff VLAN. I also left about half of the /23 unused so the firm can grow by roughly 40% without readdressing anything.

2. Client Requirements

The firm handles sensitive client material, so I treated traffic separation as a real requirement and not just nice clean design. Staff, management, voice, servers and network management each need their own space, they need to reach each other where it makes sense, and the whole thing has to be testable in Packet Tracer. My assigned challenge was 802.1Q VLAN trunking, which is why the design has more than one switch.

The brief has ten functional requirements:
(1) Reliable connectivity to network services
(2) Use of 172.30.54.0/23
(3) VLAN separation
(4) 802.1Q trunking between switches
(5) Routing between VLANs
(6) A dedicated voice network for CR11
(7) Management addressing for infrastructure
(8) Enough address space for growth
(9) A design that can be built in Packet Tracer
(10) Testing that proves it works

The design constraints were to stay specific to this client, plan for about 40% user growth over three years, and provide evidence that it works.

3. Network Design

3.1 Physical topology

R1 connects to the core switch Sw1, and Sw1 connects to two access switches, Sw2 and Sw3. Sw2 serves PC1, PC2 and IP Phone 1. Sw3 serves PC3, PC4, IP Phone 2 and Server1. All cabling is copper straight-through. Figure 1 in Appendix C shows the topology as built.

Device	Qty	Role
Cisco 1941 router (R1)	1	Inter-VLAN routing and default gateway
Cisco 2960 switch (Sw1)	1	Core / distribution
Cisco 2960 switches (Sw2, Sw3)	2	Access layer
PCs	4 (+1 added in Milestone 2)	End-user devices
Cisco 7960 IP phones	2	VoIP handsets
Server	1	Network services

Link	Type
R1 G0/0 to Sw1 G0/1	802.1Q trunk
Sw1 G0/2 to Sw2 G0/1	802.1Q trunk
Sw1 Fa0/24 to Sw3 G0/1	802.1Q trunk
Sw2 to PC1, PC2, IP Phone 1	Access
Sw3 to PC3, PC4, IP Phone 2, Server1	Access

I made one change to the topology in Milestone 2. The Settings button on the Packet Tracer 7960 phone does not respond, so I could not give the phone itself a static IP. To get fix this I attached an extra PC ("IP Phone 2 PC") to the phone's PC port and put the assigned address on that PC instead.

3.2 Logical topology

The logical layout follows the physical one. VLANs keep the traffic types apart, and the trunks carry all of them over the same cables. R1 has one sub-interface per VLAN on its link to Sw1, which is what provides the routing in FR5. I set VLAN 99 as the native VLAN on every trunk so untagged traffic does not sit in VLAN 1, which is the default and a known weak spot.

VLAN	Name	Purpose
10	STAFF-DATA	General staff data
20	MANAGEMENT	Management users and devices
30	VOICE	VoIP handsets and voice traffic (CR11)
40	SERVERS	Servers and network services
99	NETWORK-MGMT	Switch and router management; native VLAN

4. IP Addressing Plan

The assigned block is 172.30.54.0/23 (mask 255.255.254.0), which gives 512 addresses from 172.30.54.0 to 172.30.55.255. I used VLSM and carved the subnets from largest to smallest so every one starts on a clean boundary and is very close to the size it actually needs.

VLAN	Name	Network	Mask	Usable range	Gateway	R1 sub-int
10	STAFF-DATA	172.30.54.0/25	255.255.255.128	.1 - .126	172.30.54.1	g0/0.10
30	VOICE	172.30.54.128/26	255.255.255.192	.129 - .190	172.30.54.129	g0/0.30

20	MANAGEMENT	172.30.54.192/27	255.255.255.224	.193 - .222	172.30.54.193	g0/0.20
40	SERVERS	172.30.54.224/28	255.255.255.240	.225 - .238	172.30.54.225	g0/0.40
99	NETWORK-MGMT	172.30.55.0/28	255.255.255.240	172.30.55.1 - .14	172.30.55.1	g0/0.99

Each gateway is the first address in its subnet, so hosts start at the second address. Infrastructure devices use static addresses: Sw1 172.30.55.2, Sw2 172.30.55.3 and Sw3 172.30.55.4 in VLAN 99, and Server1 172.30.54.226 in VLAN 40. For everything else I planned DHCP ranges of 172.30.54.10 to .126 for staff, 172.30.54.140 to .190 for voice and 172.30.54.200 to .222 for management. A staff PC's ipconfig output lists 172.30.54.1 (R1) as its DHCP server, and the voice-range address 172.30.54.140 answered my ping.

Planning for growth. I assumed starting sizes of 60 staff, 15 management users, 20 phones and 8 servers, then added 40%. Every subnet still has room:

Segment	Initial	+40%	Required	Subnet capacity
Staff	60	24	84	126
Management	15	6	21	30
Voice	20	8	28	62
Servers	8	4	12	14

The five subnets use 128 + 64 + 32 + 16 + 16 = 256 addresses. The other 256 (172.30.54.240 to .255 and 172.30.55.16 to .255) stay free for things like extra staff VLANs, wireless, a guest network, more phones and new departments.

5. Implementation

5.1 Router R1: router-on-a-stick

These are the sub-interfaces as they appear in R1's running configuration (Figures 8 and 9). The physical interface carries no address and is brought up with `no shutdown`.

```
interface GigabitEthernet0/0.10
encapsulation dot1Q 10  ip address
172.30.54.1 255.255.255.128 interface
GigabitEthernet0/0.20  encapsulation
dot1Q 20  ip address 172.30.54.193
255.255.255.224 interface
GigabitEthernet0/0.30  encapsulation
dot1Q 30  ip address 172.30.54.129
255.255.255.192 interface
GigabitEthernet0/0.40  encapsulation
dot1Q 40  ip address 172.30.54.225
255.255.255.240 interface
GigabitEthernet0/0.99  encapsulation
dot1Q 99 native  ip address 172.30.55.1
255.255.255.240
```

5.2 Core switch Sw1: VLANs and trunks

```
vlan 10  name
STAFF-DATA vlan
20  name
MANAGEMENT vlan
30  name VOICE
vlan 40  name
SERVERS vlan 99
name NETWORK-MGMT
interface range Fa0/24, Gig0/1, Gig0/2
switchport mode trunk  switchport trunk
native vlan 99  switchport trunk allowed
vlan 10,20,30,40,99
```

`show interfaces trunk` on Sw1 (Figure 10) shows Fa0/24, Gig0/1 and Gig0/2 trunking with 802.1q and native VLAN 99, with VLANs 10, 20, 30, 40 and 99 allowed.

5.3 Access switches Sw2 and Sw3

Sw2 and Sw3 carry VLANs 10, 20, 30 and 40 and are trunked back to Sw1. The ports that connect IP phones use a data VLAN plus the voice VLAN, so the phone's voice traffic is tagged into VLAN 30:

```
interface FastEthernet0/3
switchport mode access
switchport access vlan 10
switchport voice vlan 30
```

Management addresses for the switches belong in VLAN 99, for example `interface Vlan99` with `ip address 172.30.55.4 255.255.255.240` on Sw3 and `ip default-gateway 172.30.55.1`.

6. Assigned Feature: CR11 VoIP, Static IP 172.30.54.130

My assigned feature is a static address of 172.30.54.130 inside the Voice VLAN. It sits in VLAN 30 (172.30.54.128/26), uses the mask 255.255.255.192 and the gateway 172.30.54.129 on R1's g0/0.30 sub-interface, and points to Server1 (172.30.54.226) for DNS. Figure 2 shows the configuration.

Item	Value
Network	VLAN 30 VOICE, 172.30.54.128/26
Assigned IP	172.30.54.130
Subnet mask	255.255.255.192
Default gateway	172.30.54.129 (R1 g0/0.30)
DNS server	172.30.54.226
Device	Host attached to IP Phone 2's PC port on Sw3
Port configuration	switchport access vlan 10, switchport voice vlan 30

The catch is the Packet Tracer limitation I mentioned earlier: the 7960's Settings button does not respond, so the address cannot be set on the phone. I put it on the PC attached to IP Phone 2 instead, in the Voice VLAN and using the Voice VLAN gateway. This is a workaround, not a perfect model of a real handset, but it still shows the dedicated voice network CR11 and FR6 ask for, and it lets the assigned address be tested end to end.

7. Verification and Testing

I ran the connectivity tests from PC4, which sits in the staff VLAN, and checked the router and Sw1 with show commands. Screenshots are in Appendix D to F.

#	Test	From / to	Expected	Actual	Result
1	Inter-VLAN to assigned IP	PC4 (VLAN 10) to 172.30.54.130 (VLAN 30)	4/4 replies	4/4, 0% loss, TTL 127	PASS
2	Gateway, VLAN 10	PC4 to 172.30.54.1	4/4 replies	4/4, TTL 255	PASS
3	Gateway, VLAN 30	PC4 to 172.30.54.129	4/4 replies	4/4, TTL 255	PASS
4	Gateway, VLAN 20	PC4 to 172.30.54.193	4/4 replies	4/4, TTL 255	PASS
5	Gateway, VLAN 40	PC4 to 172.30.54.225	4/4 replies	4/4, TTL 255	PASS
6	Voice VLAN device	PC4 to 172.30.54.140	4/4 replies	4/4, TTL 254	PASS
7	VLANs and trunks	Sw1: show vlan brief, show interfaces trunk	VLANs 10,20,30,40,99 present; trunks up, native 99	As expected	PASS
8	Router interfaces	R1: show ip interface brief	All sub-interfaces up/up	g0/0.10, .20, .30, .40, .99 up/up	PASS

The TTL values matter. A reply with TTL 127 or 254 has crossed the router, which is what proves inter-VLAN routing and the trunk are working. The first attempt to ping 172.30.54.130 timed out (100% loss) and the next attempt succeeded. That is normal while ARP entries are still being learned, and every later test replied straight away.

8. Challenges and Known Issues

IP phone configuration. The 7960 phone cannot be configured from the Packet Tracer GUI, so I moved the assigned address onto an attached PC, as described in section 6.

Inconsistent VLAN numbers. In my Milestone 2 table, VLAN 20 was labelled VOICE and VLAN 30 MANAGEMENT, which is the wrong way round compared with Milestone 1 and with the actual configuration. This report follows Milestone 1: VLAN 20 is MANAGEMENT and VLAN 30 is VOICE.

VLAN names on Sw3. When I ran show vlan brief on Sw3, the VLANs showed default names such as VLAN0010. The fix is to add `name` commands under each VLAN on the access switches so they match Sw1.

VLAN 99 on the access switches. On Sw1, VLAN 99 is forwarding on Gig0/1 and Fa0/24 but not on Gig0/2, which suggests VLAN 99 was never created on Sw2. Creating VLAN 99 (NETWORK-MGMT) on Sw2 and Sw3 makes the native VLAN and the switch management addresses consistent across the whole network.

9. Conclusion and Recommendations

The network does what the brief asked. It separates traffic into five VLANs, carries them over 802.1Q trunks with a non-default native VLAN, routes between them on R1, gives voice its own VLAN for CR11 with the assigned address 172.30.54.130, and has enough spare address space for 40% growth. The tests in section 7 back each of those claims up.

If the firm took this further, the next steps I would suggest are: formal DHCP pools or a relay for each VLAN, port security on the access ports, SSH limited to VLAN 99, QoS to prioritise voice traffic in VLAN

30, a redundant uplink from Sw1 to each access switch with spanning tree tuned for it, wireless and guest VLANs using the spare space from 172.30.55.16, and ACLs on R1 to restrict access to the SERVERS and NETWORK-MGMT VLANs.

10. Github repository

https://github.com/shaun101-mol/CMPG-325-MS-MOLEWA

Appendix A to C: Topology and addressing

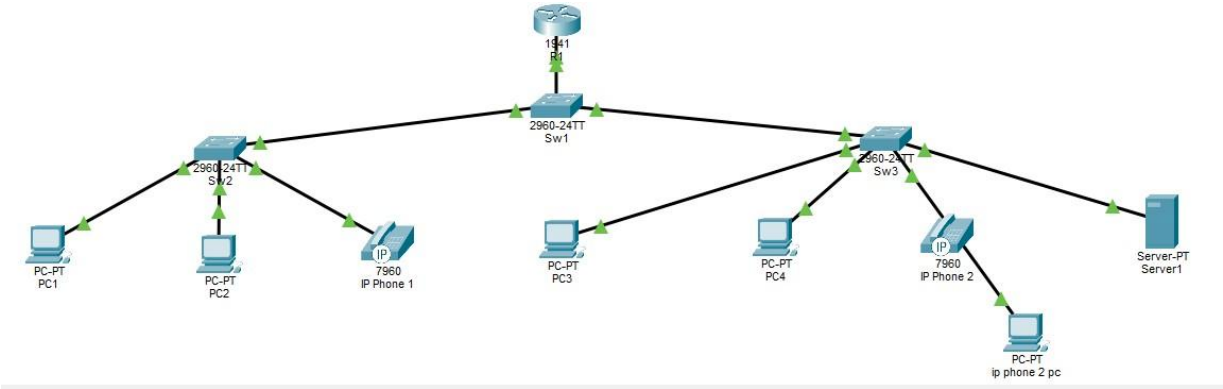


Figure 1: Packet Tracer topology (Appendix C), including the extra PC on IP Phone 2.

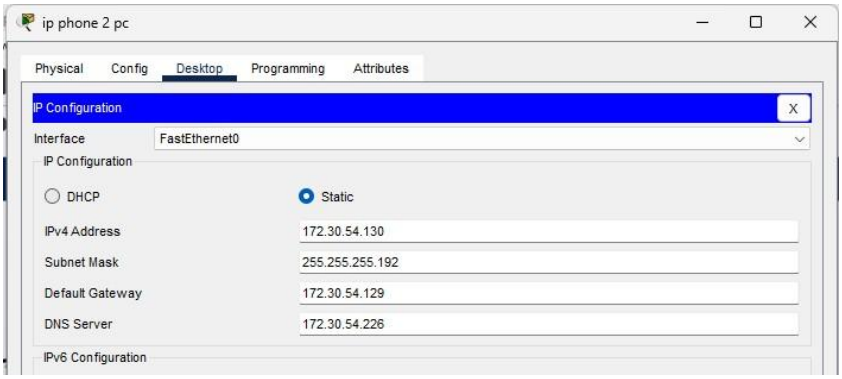


Figure 2: Static IP configuration of the host on IP Phone 2's PC port: 172.30.54.130, mask 255.255.255.192, gateway 172.30.54.129.

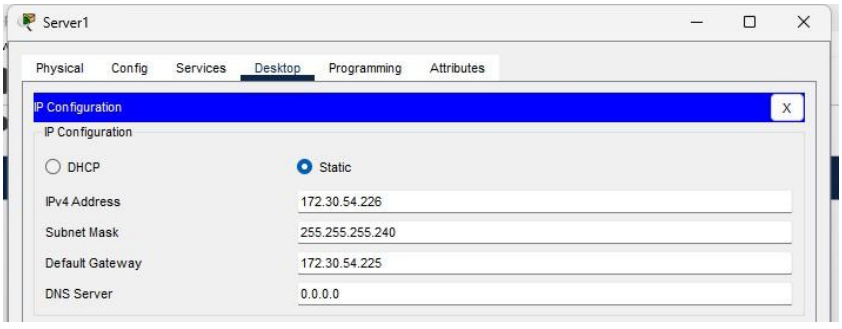


Figure 3: Server1 static configuration: 172.30.54.226, mask 255.255.255.240, gateway 172.30.54.225.

Appendix D: Ping tests from PC4 (staff VLAN)

```
C:\>ping 172.30.54.130

Pinging 172.30.54.130 with 32 bytes of data:

Reply from 172.30.54.130: bytes=32 time<1ms TTL=127
Reply from 172.30.54.130: bytes=32 time=10ms TTL=127
Reply from 172.30.54.130: bytes=32 time=2ms TTL=127
Reply from 172.30.54.130: bytes=32 time<1ms TTL=127

Ping statistics for 172.30.54.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 3ms

C:\>
```

Figure 4: PC4 to 172.30.54.130 (assigned Voice VLAN address): 4/4 replies, TTL 127.

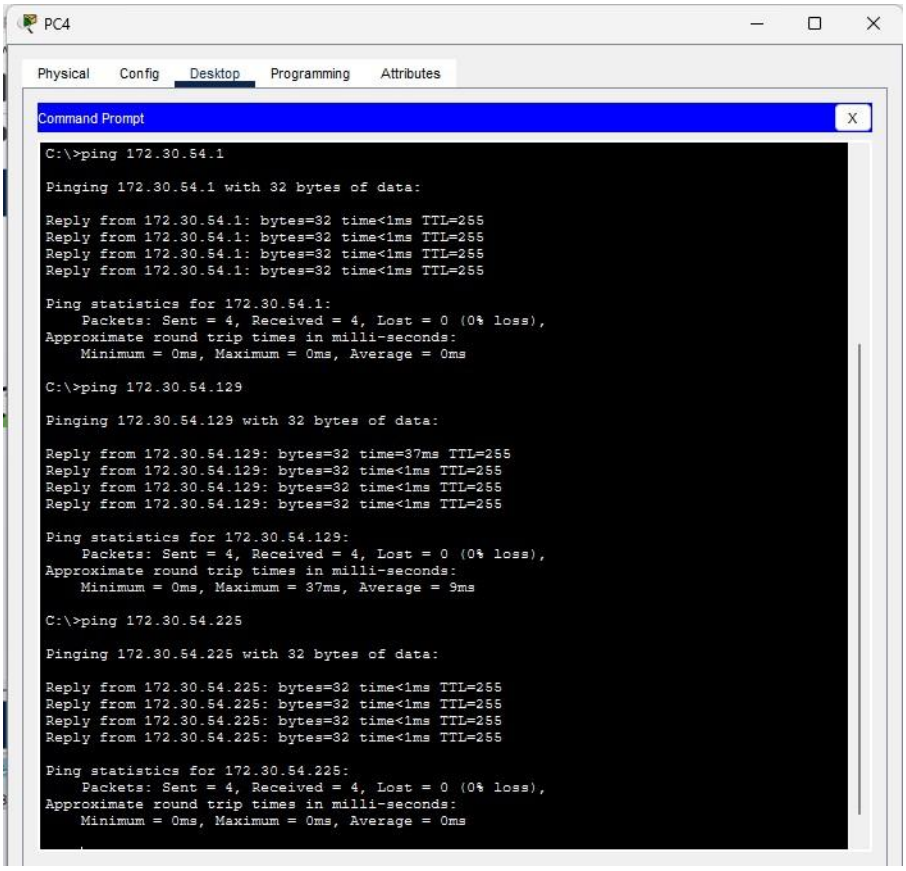


Figure 5: PC4 to the gateways 172.30.54.1, 172.30.54.129 and 172.30.54.225.

```
C:\>ping 172.30.54.193

Pinging 172.30.54.193 with 32 bytes of data:

Reply from 172.30.54.193: bytes=32 time<1ms TTL=255
Reply from 172.30.54.193: bytes=32 time<1ms TTL=255
Reply from 172.30.54.193: bytes=32 time<1ms TTL=255
Reply from 172.30.54.193: bytes=32 time<1ms TTL=255

Ping statistics for 172.30.54.193:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 6: PC4 to the management gateway 172.30.54.193.

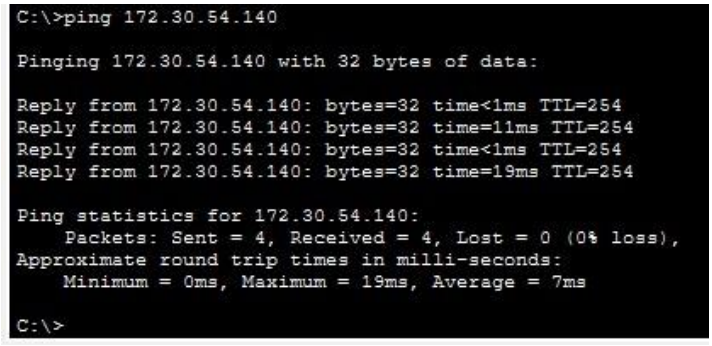
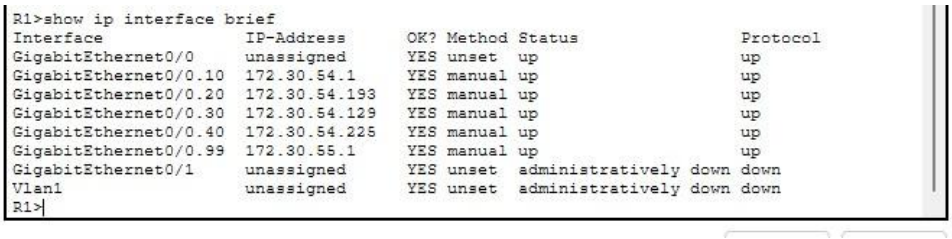


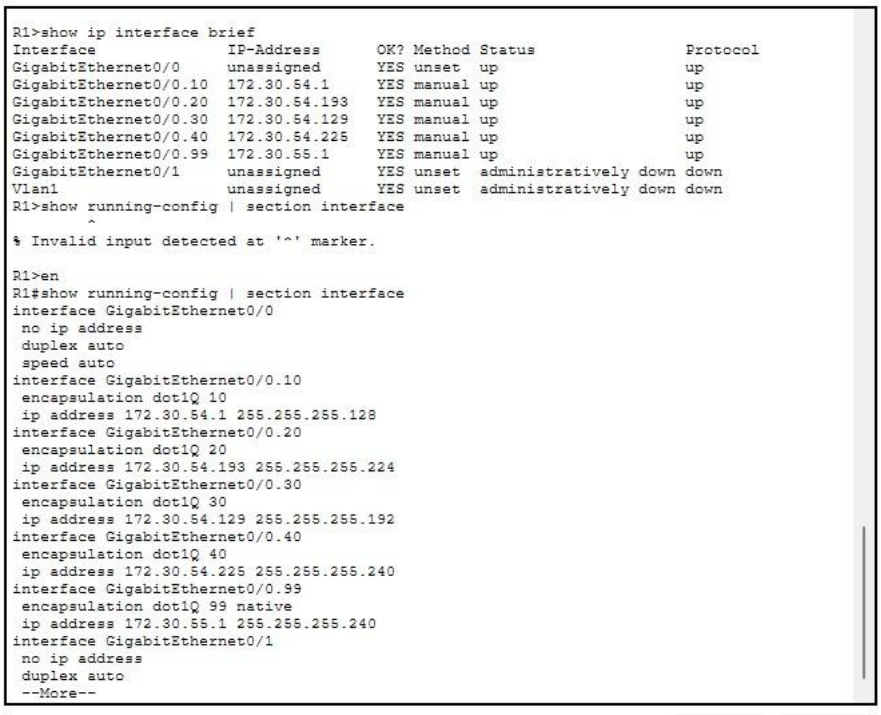
Figure 7: PC4 to 172.30.54.140 in the voice DHCP range: 4/4 replies, TTL 254.

Appendix E: Router verification



```
R1>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  172.30.54.1     YES manual up                    up
GigabitEthernet0/0.20  172.30.54.193   YES manual up                    up
GigabitEthernet0/0.30  172.30.54.129   YES manual up                    up
GigabitEthernet0/0.40  172.30.54.225   YES manual up                    up
GigabitEthernet0/0.99  172.30.55.1     YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1>
```

Figure 8: R1 show ip interface brief: all sub-interfaces up/up.



```
R1>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  172.30.54.1     YES manual up                    up
GigabitEthernet0/0.20  172.30.54.193   YES manual up                    up
GigabitEthernet0/0.30  172.30.54.129   YES manual up                    up
GigabitEthernet0/0.40  172.30.54.225   YES manual up                    up
GigabitEthernet0/0.99  172.30.55.1     YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1>show running-config | section interface
        ^
% Invalid input detected at '^' marker.

R1>en
R1#show running-config | section interface
interface GigabitEthernet0/0
 no ip address
 duplex auto
 speed auto
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 172.30.54.1 255.255.255.128
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 172.30.54.193 255.255.255.224
interface GigabitEthernet0/0.30
 encapsulation dot1Q 30
 ip address 172.30.54.129 255.255.255.192
interface GigabitEthernet0/0.40
 encapsulation dot1Q 40
 ip address 172.30.54.225 255.255.255.240
interface GigabitEthernet0/0.99
 encapsulation dot1Q 99 native
 ip address 172.30.55.1 255.255.255.240
interface GigabitEthernet0/1
 no ip address
 duplex auto
 --More--
```

Figure 9: R1 running configuration for the sub-interfaces.

Appendix F: Switch verification

IOS Command Line Interface

```
Sw1>show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23
10   STAFF-DATA                       active
20   MANAGEMENT                       active
30   VOICE                            active
40   SERVERS                          active
99   NETWORK-MGMT                     active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Sw1>show int trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/24      on           802.1q         trunking      99
Gig0/1      on           802.1q         trunking      99
Gig0/2      on           802.1q         trunking      99

Port        Vlans allowed on trunk
Fa0/24      10,20,30,40,99
Gig0/1      10,20,30,40,99
Gig0/2      10,20,30,40,99

Port        Vlans allowed and active in management domain
Fa0/24      10,20,30,40,99
Gig0/1      10,20,30,40,99
Gig0/2      10,20,30,40,99

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/24      10,20,30,40,99
Gig0/1      10,20,30,40,99
Gig0/2      10,20,30,40
```

Figure 10: Sw1 show vlan brief and show interfaces trunk.